

Phylogenetic signal of KO presence — all curated metal KOs (n=276)

--- Niche breadth $\lambda = 0.766$

Pagel's λ (gene presence fraction, intercept-only)

1.0
0.8
0.6
0.4
0.2
0.0

Resistance/Detoxification
(n=59)

Cofactor Biosynthesis
(n=4)

Sensing/Regulation
(n=22)

Transport/Homeostasis
(n=55)

Metal-dependent Metabolism
(n=15)

Unknown
(n=121)

